Uninstallable by Design

The Role of Pre-installed Apps in Android's Security Landscape

Thomas Sutter, July 2025

Who am I

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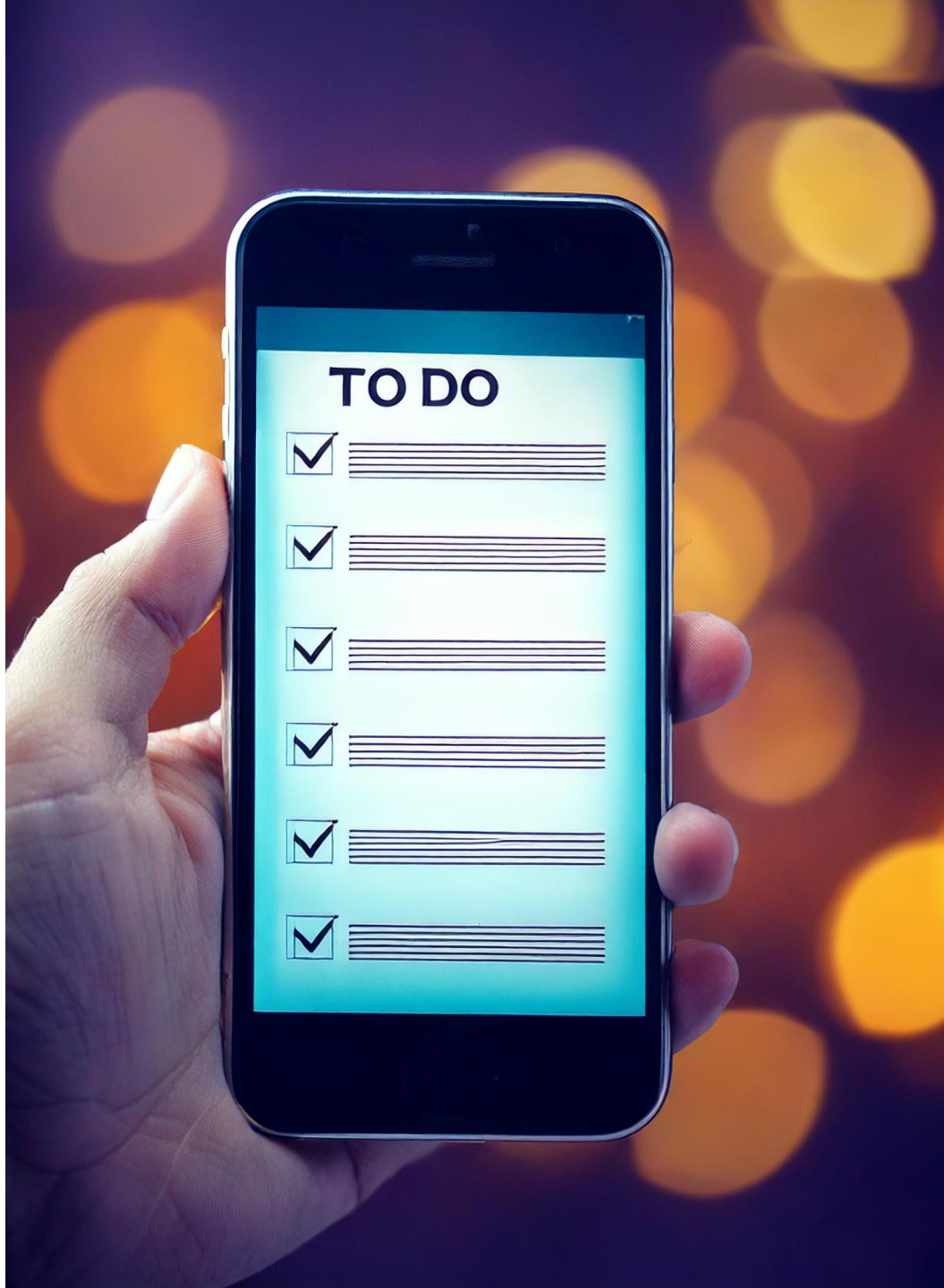
Content

- What are pre-installed apps?
- How Android secure boot works
- Why we can’t uninstall pre-installed apps

- How to test and regain some transparency



What are pre-installed apps?

Different types of apps

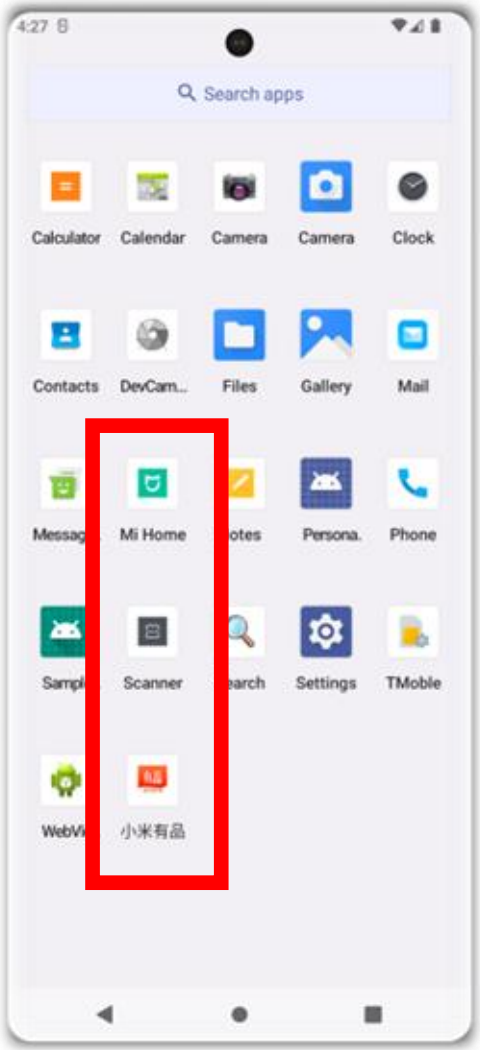
Userspace Apps

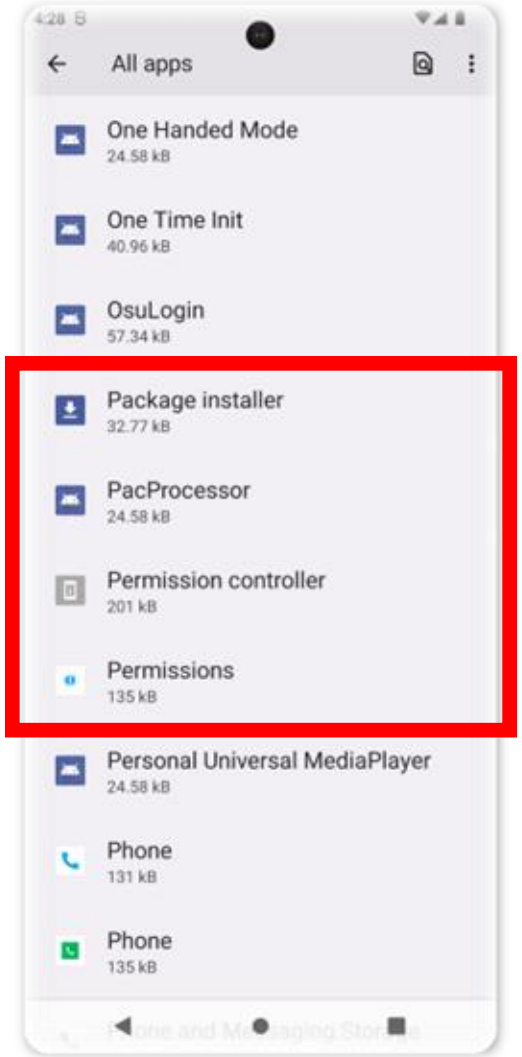
- Apps installed over an app store
- User reviews available
- Users can decice to uninstall apps
- Apps have limited access rights

Different types of apps

Userspace Apps

Pre-installed Apps





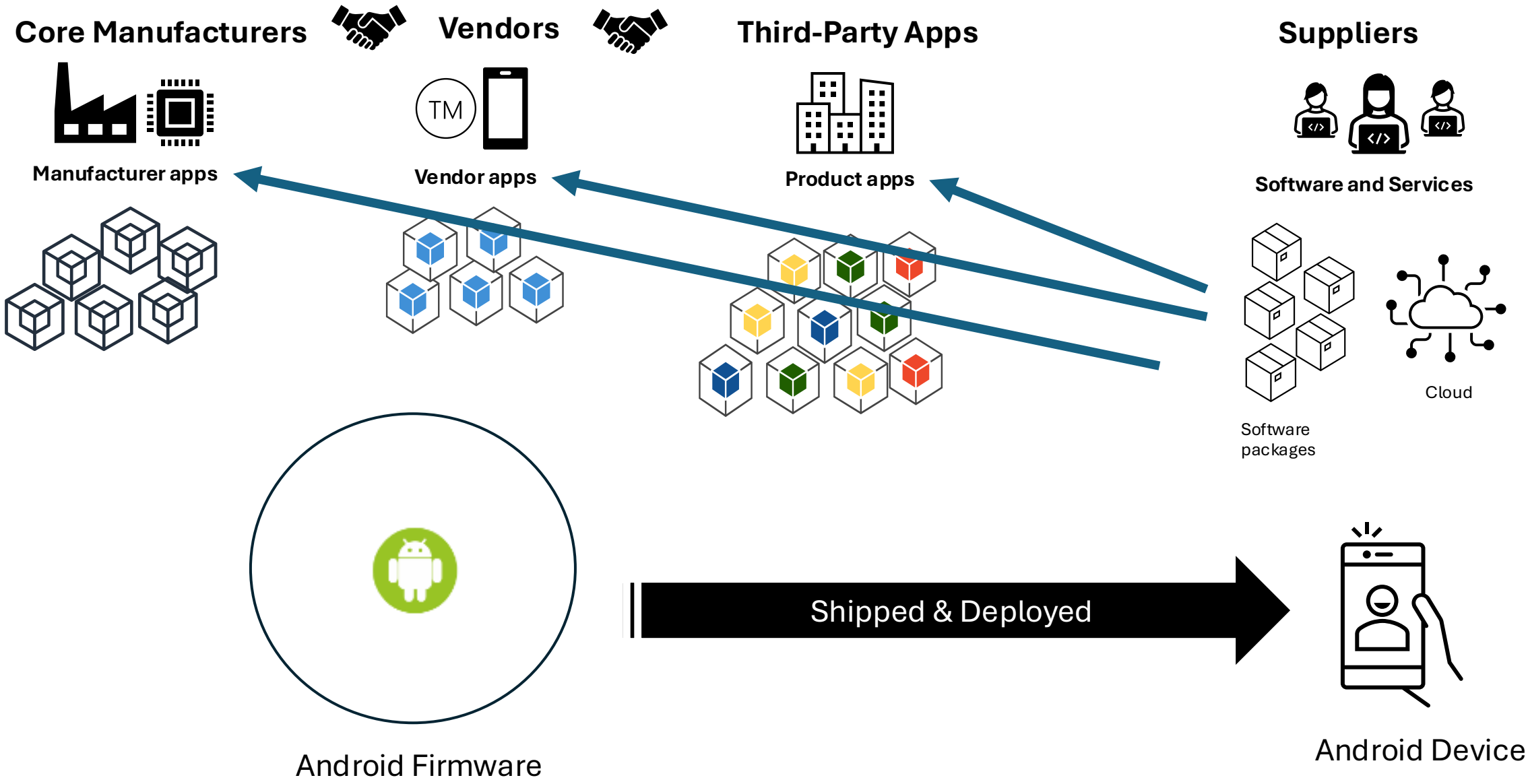
Pre-installed apps are all the Android applications that are shipped with a device

For instance:

- Telephone app
- Contacts app
- System service
  - Headless apps
- ...

Apps are stored in the “.apk” file format.

Android Supply Chain

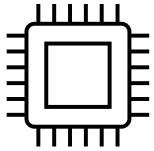


About Android Security

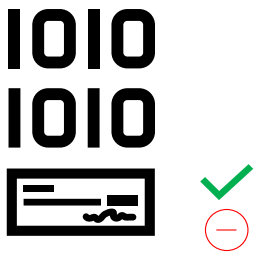
Android Verified Boot (simplified)

Hardware Root of Trust

Crypto Chip (SE)



Bootloaders (ROM + Flash)



ROM part verifies flash bootloader (updates)
Flash part verifies kernel
Verification with the "Bootloader key"

Linux Kernel

Kernel

Partition Signature (Merkle Tree)

VBMeta

VBMeta includes the cryptographic meta-data (public keys) from different sources (Vendors, ODM, ...)

Device partitions

System	Read
Partition Signature (Merkle Tree)	
User	Read + Write
Vendor	Read
Partition Signature (Merkle Tree)	
Product	Read
Partition Signature (Merkle Tree)	
...	Read
Partition Signature (Merkle Tree)	

"Unchangable" (on chip):

- Bootloader key (asymmetric)

"Chain of Trust" to protect the integrity of files

Verification is done before mount and during
FS-verity for specific files runtime

Programmable:

- Android Root of Trust (asymmetric)

Android follows a defense in-depth approach by using AVB in combination with SELinux

Why are partitions read-only on Android?

Minimizing attack surface during runtime

Sandboxing

Performance

Software Attestation

Files on read-only partitions cannot simply be deleted

Consequenlty, pre-installed apps cannot be simply uninstalled

- Users have no way to remove these apps
  - Only by jailbreaking or rooting, which is not a real option for normal users

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  - No review possibility
  - No publicly known security controls
  - No app classification or description
  - Apps cannot easily be installed

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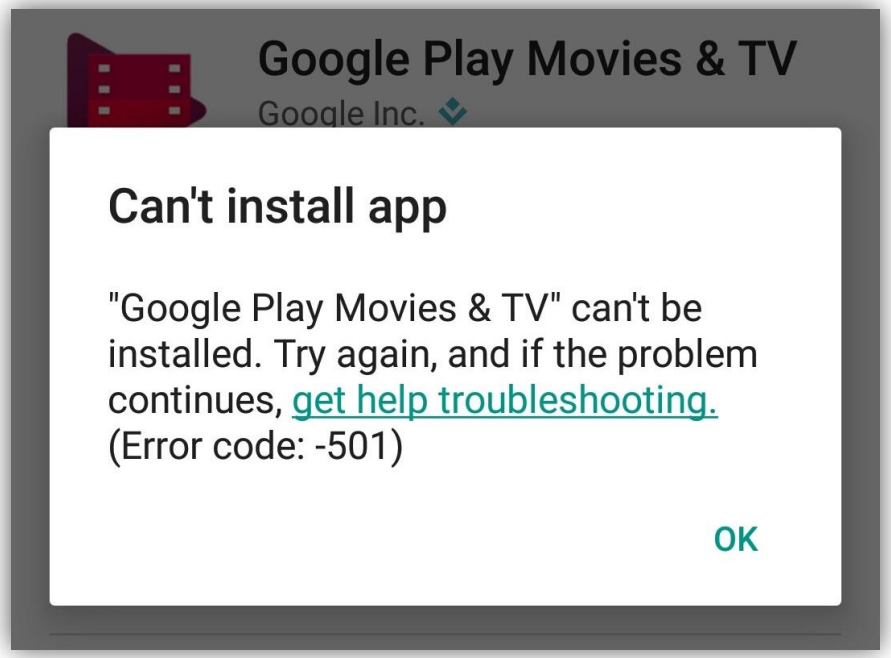
- Manufacturers and vendors sell space on their products



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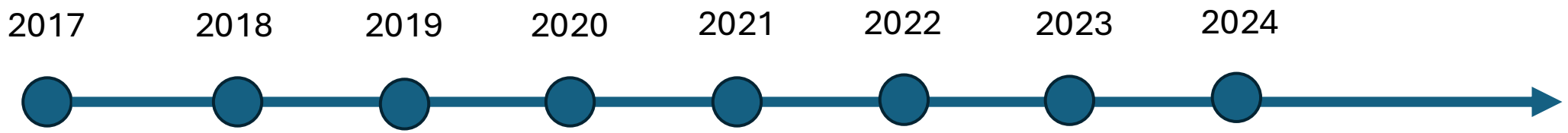
- Pre-installed app have lots of permissions
- Manufacturers and vendors sell space on their products
- Security practioners have problems analysing them on scale and in-depth
  - Installation is not simply possible
  - Dependencies to files and services

History of Incidents



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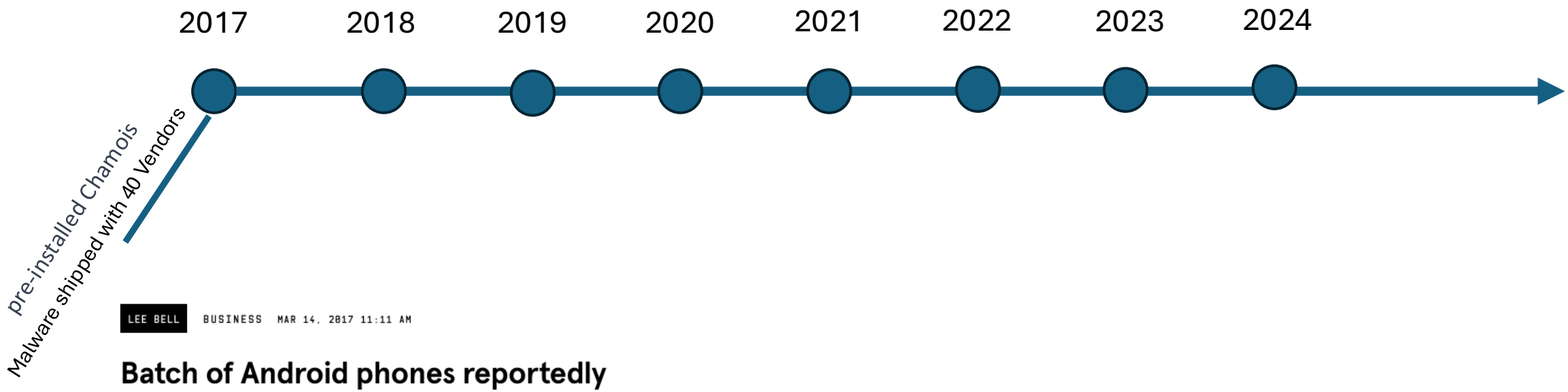


LEE BELL BUSINESS MAR 14, 2017 11:11 AM

Batch of Android phones reportedly shipped with malware pre-installed

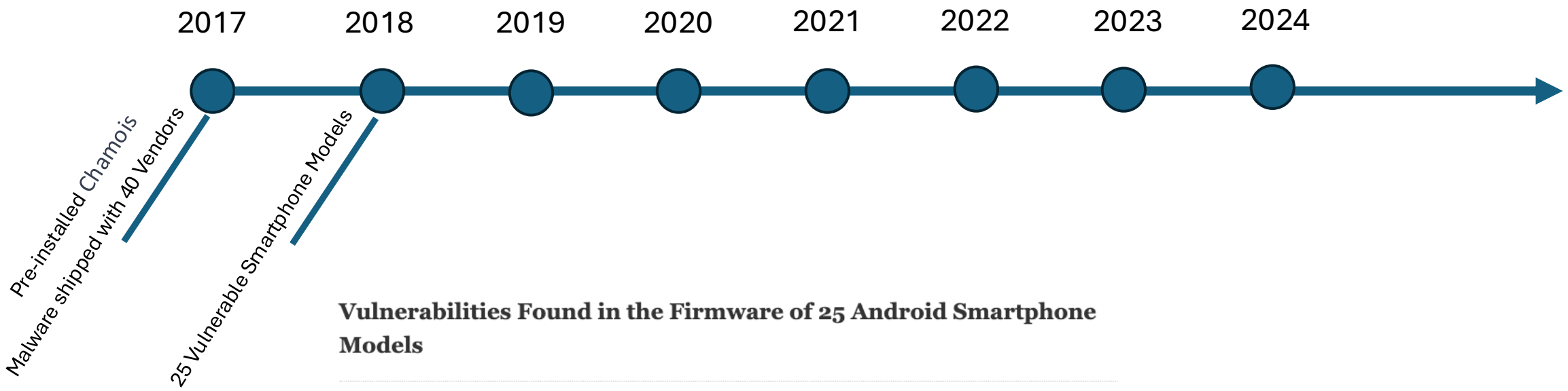
The likes of Samsung and Lenovo are included in the list of affected handsets, according to security experts

ALMOST 40 DIFFERENT Android smartphones are shipping infected with malware that has been installed during the supply chain and manufacturing process, according to analysis by security experts.

Check Point Security claims the 38 phones affected aren't from little-known manufacturers but from big names such as Samsung and Lenovo. Research by the company found "a severe infection" in 38 Android devices, belonging to a large telecommunications company and a multinational technology company," said Oren Koriat of the Check Point mobile research team.

https://www.wired.com/story/android-phones-hiding-pre-installed-malware/

History of Incidents



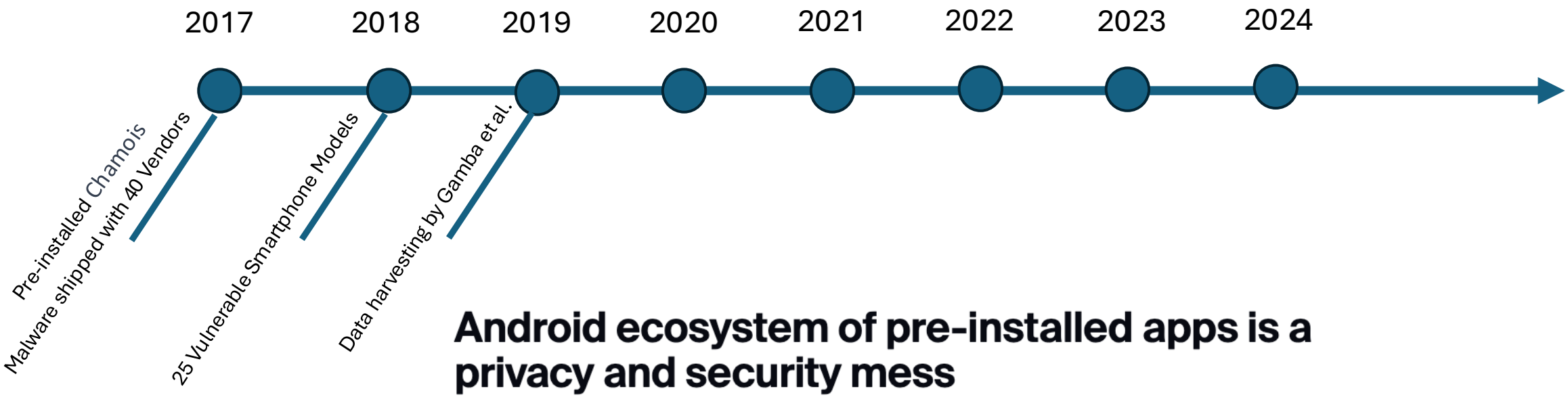
Vulnerabilities Found in the Firmware of 25 Android Smartphone Models

By **Catalin Cimpanu** August 12, 2018 06:11 PM 0

These vulnerabilities were discovered in both the default apps that come preinstalled on some devices by default (and are sometimes unremovable), but also in the firmware of core device drivers that can't be removed without losing some of the phone's functionality, if not access to the device as a whole.

https://www.bleepingcomputer.com/news/security/vulnerabilities-found-in-the-firmware-of-25-android-smartphone-models/

History of Incidents



Android ecosystem of pre-installed apps is a privacy and security mess

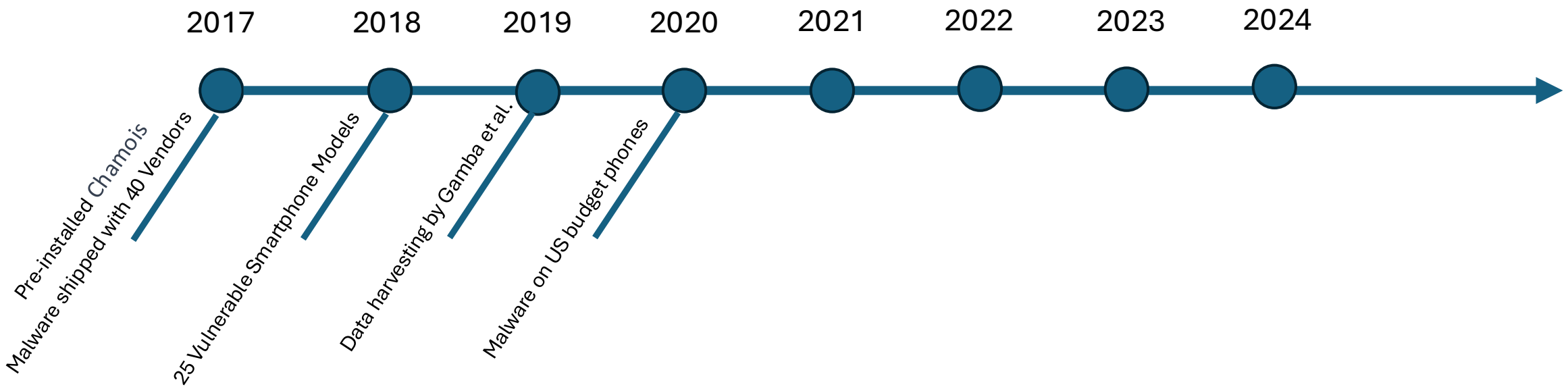
Extensive academic study finds data-harvesting and malware-laced pre-installed apps.

Written by **Catalin Cimpanu,** Contributor
March 25, 2019 at 3:05 p.m. PT

https://www.zdnet.com/article/android-ecosystem-of-pre-installed-apps-is-a-privacy-and-security-mess/

Paper: “An Analysis of Pre-installed Android Software” by Gamba et al.

History of Incidents



More pre-installed malware has been found in budget US smartphones

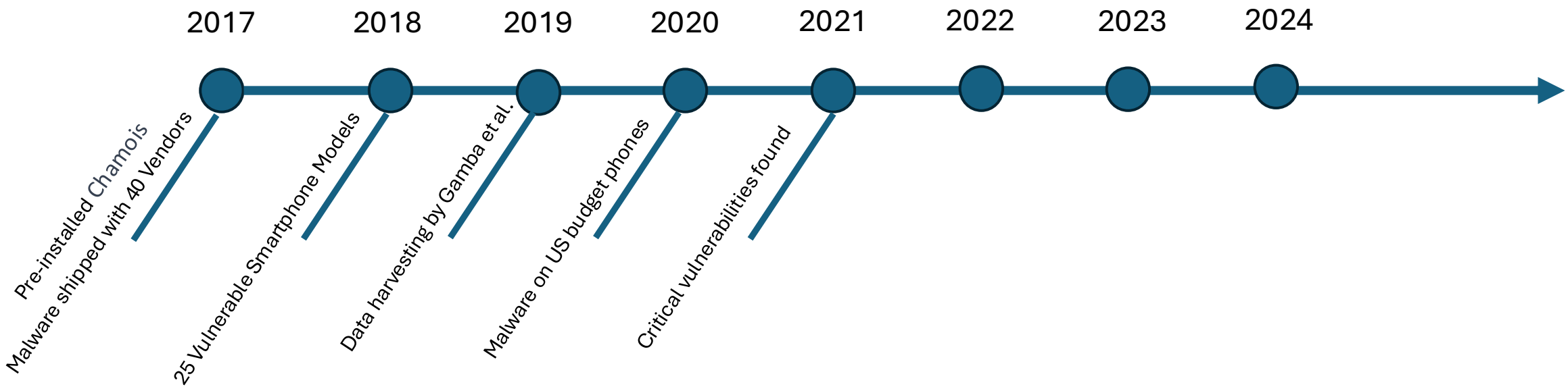
Cheap phones often have tradeoffs but researchers say this should never compromise user safety.

Written by **Charlie Osborne,** Contributing Writer

July 8, 2020 at 9:40 p.m. PT

https://www.zdnet.com/article/more-pre-installed-malware-has-been-found-in-budget-us-smartphones/

History of Incidents



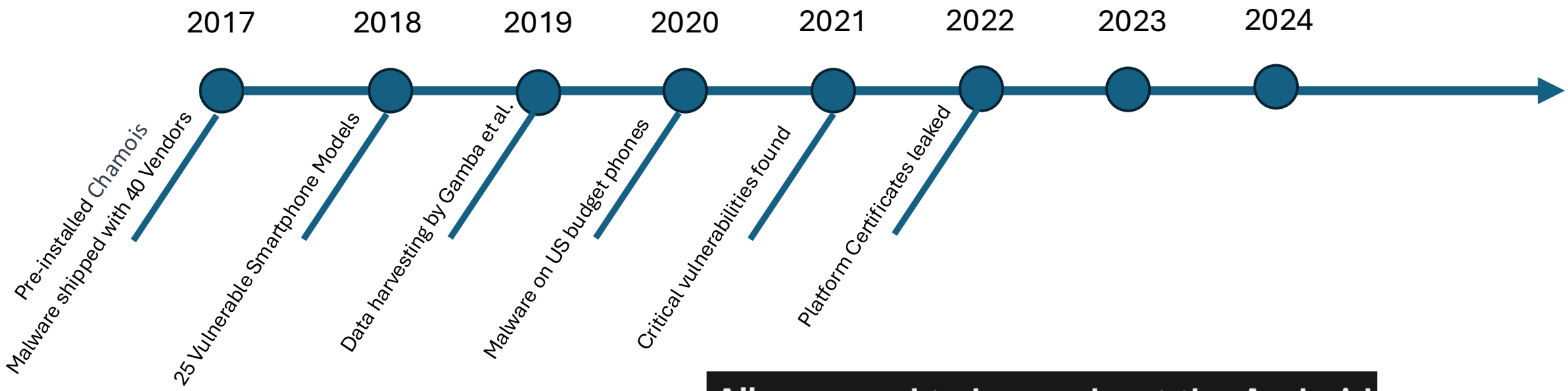
Hackers Can Exploit Samsung Pre-Installed Apps to Spy On Users

Jun 11, 2021 Ravie Lakshmanan

Multiple critical security flaws have been disclosed in Samsung's pre-installed Android apps, which, if successfully exploited, could have allowed adversaries access to personal data without users' consent and take control of the devices.

https://thehackernews.com/2021/06/hackers-can-exploit-samsung-pre.html

History of Incidents



All you need to know about the Android platform certificate leak

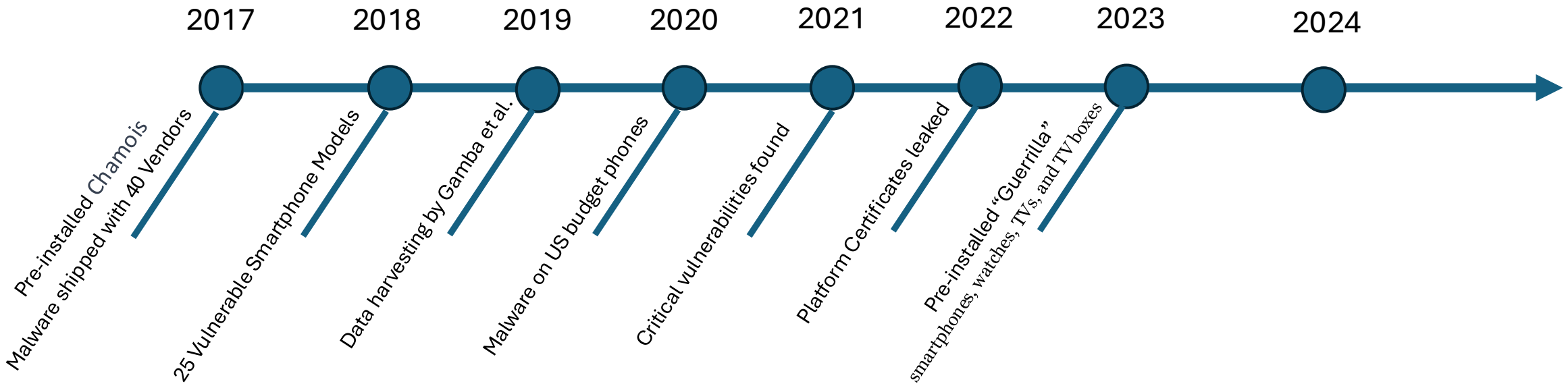
Rest assured that it can no longer be exploited

BY MANUEL VONAU PUBLISHED DEC 9, 2022

Google disclosed a serious issue mainly affecting Samsung phones toward the end of 2022. Some platform certificates from Samsung got into the hands of bad actors, which allowed them to create malware with elevated permissions, potentially allowing hackers to hijack phones by loading tampered software on them. This seems to affect all phones from a given manufacturer, regardless of whether you have Android 13. Here's everything we know about the vulnerability and what you can do to protect yourself and your phone.

https://www.androidpolice.com/android-platform-certificate-leak-explainer/

History of Incidents



MALWARE & THREATS

Millions of Smartphones Distributed Worldwide With Preinstalled 'Guerrilla' Malware

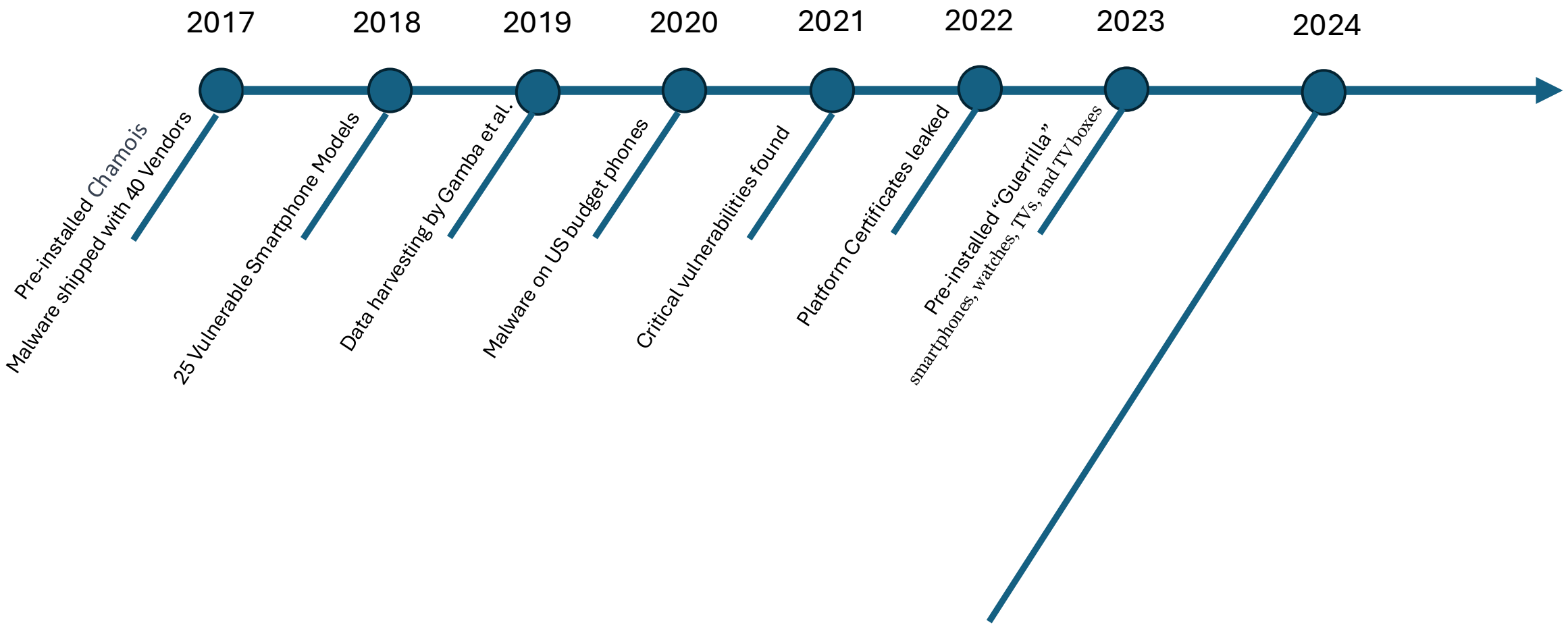
A threat actor tracked as Lemon Group has control over millions of smartphones distributed worldwide thanks to preinstalled Guerrilla malware.

By Eduard Kovacs
May 18, 2023

A threat actor has control over millions of smartphones distributed worldwide thanks to a piece of malware that has been preinstalled on the devices, Trend Micro warned.

https://www.securityweek.com/millions-of-smartphones-distributed-worldwide-with-preinstalled-guerrilla-malware/

History of Incidents



Can we improve this?

Research Motivation

Our main goal is to make the analysis less laboursome

- Are pre-installed apps secure?
  - How are they tested?
- How can we test pre-installed apps?
  - Static analysis?
  - Dynamic analysis?
  - Emulator vs Device?
- Can we compare different devices?
  - Can we measure security maturity?
  - Can we identify risks?
  - What kind of data is collected and shared?

Getting The Sample Data

Where to find the data?

- Extracting firmware/apps from devices?
  - Crowed-sourcing?

Where to find the data?

- ~~Extracting firmware/apps from devices?~~
  - ~~Crowed-sourcing?~~

- Official vendor websites?
  - Large vendors mostly don’t share their firmware
  - Google, LineageOS, GrapheneOS, etc.

Where to Firmware Samples?

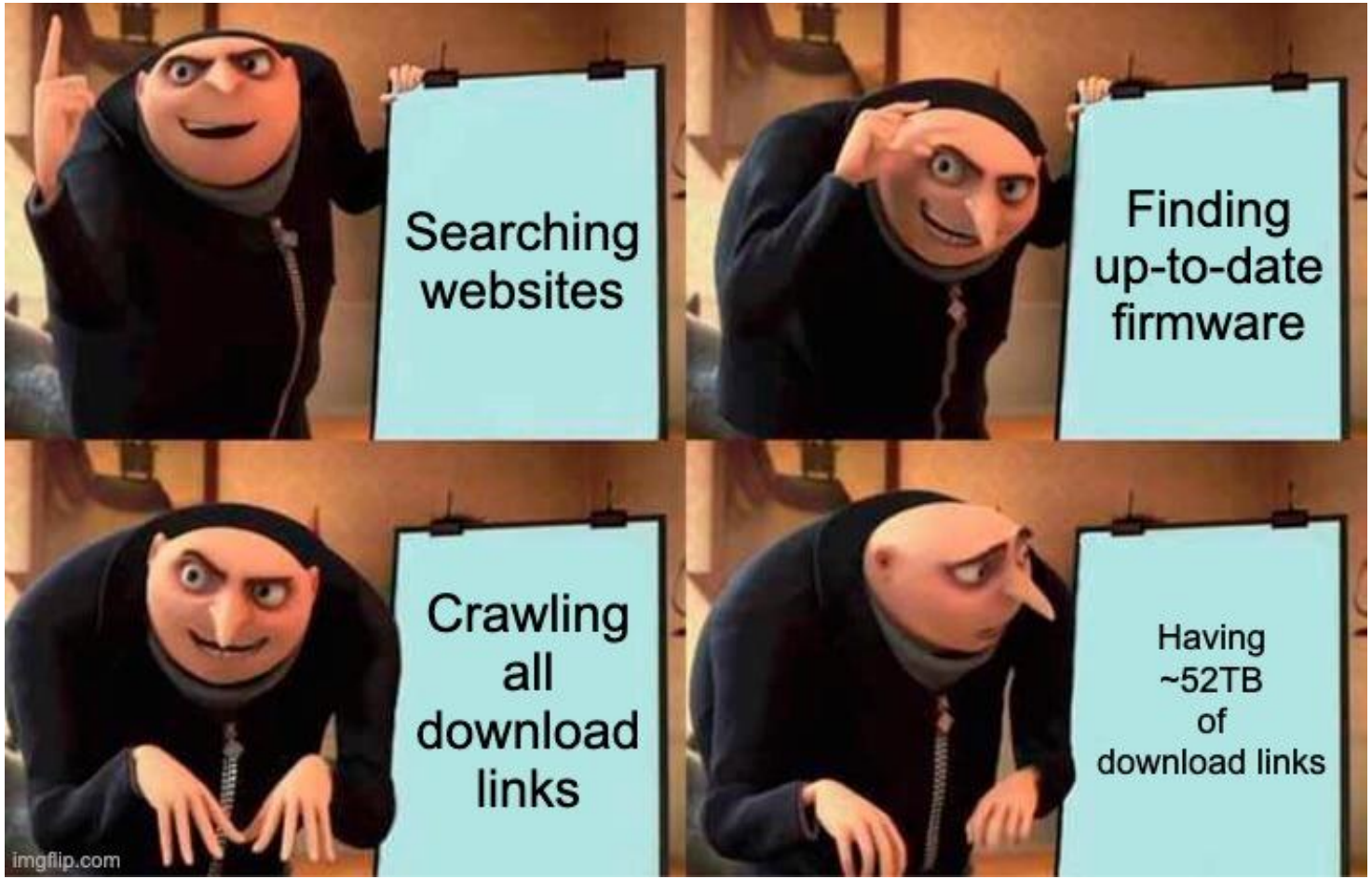
- ~~Extracting firmware/apps from devices?~~
  - ~~Crowed-sourcing?~~

- Official vendor websites?
  - Large vendors mostly don’t share their firmware
  - Google, LineageOS, GrapheneOS, etc.

- Unofficial websites sharing firmware
  - Many websites claiming to have ”official” stock firmware
    - Sometimes stock firmware is copied from official update servers
    - Hobbyists creating Android firmware
    - Fake “stock” firmware (repacking)

Crawling For Firmware

Developed a small 321 lines of code web-crawler. Just to collect some download links from various websites...



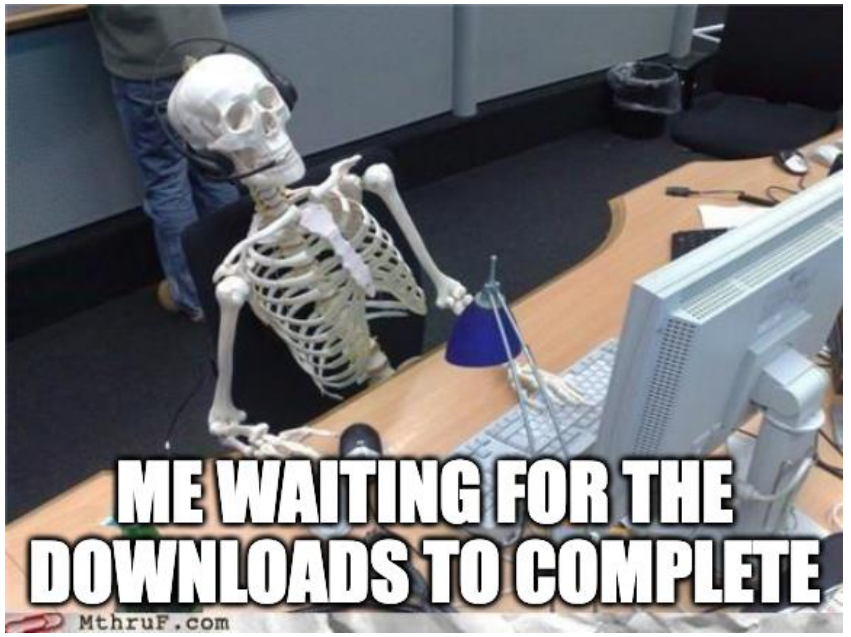
...worked better than expected

Let’s Get Research Data

2021: We collected around ~10TB of firmware samples
2023: We collected links for ~52TB and downloaded another ~16TB of samples for our current study

Basically, we found download links to all Android versions (v4 to v15)

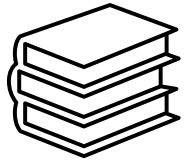
- For free
- Including official stock ROMs, fake stock ROMS, and customs ROMs



Total Bytes: 52.39 TB **Bytes Loaded:** 16.29 TB

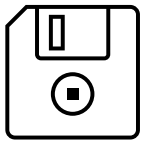
Data is available for academic researchers…
For the others: I might have a text file with some download links, that I’m eventually willing to share

A Word on File Formats



Compression Formats

.zip
.tar
.gzip
.7z
.lz4
.bin
.dat
.br
.nb0
.pac
.ubi
.ozip
...



Partition Formats

.ext2
.ext4
.sparse
.f2fs
.erofs
.payload_bin
.img
.bin
.raw
.super
...

Android container formats

.apex
.apk
.aab
...

Not including formats for kernel, radio, tos, pvmfw, cache, ...

Analysis Automation

Developing an Extraction Routine

- Maybe some of you know the tool “unblob”
  - Upsides:
    - Really handy to extract compressed files and some partitions
    - Supports Android sparse images
  - Downsides:
    - Performance is often slow in case of large firmware
    - Android custom firmware formats not supported
    - Extraction depth is hard to set
    - Came out years after me implementing an extraction routine

- Developed an own extraction routine with a mix of state-of-the-art tools and mounting tools to handle Android’s mix of file formats
  - 2021: We were able to extract ~8’000 samples
  - 2024: We were able to extract ~20k samples
    - Still open challenges in doing this stable, generic, and fast

FIRMWAREDROID

Code: https://github.com/FirmwareDroid/FirmwareDroid

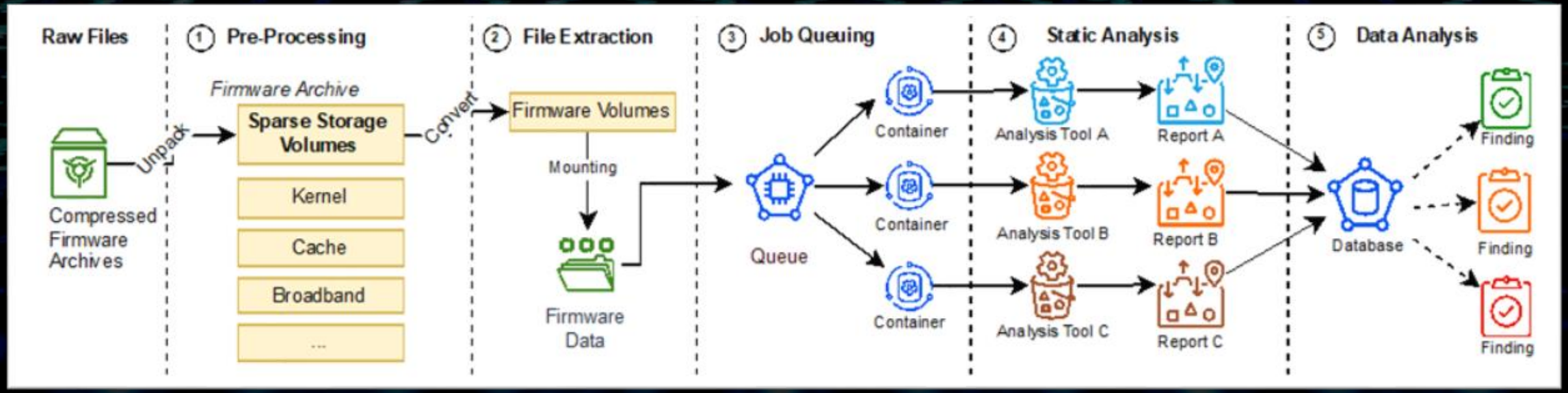


Figure from our paper: https://ieeexplore.ieee.org/document/10172951

FMD Features

- Firmware unpacking
  - Extracting files from all file partitions (e.g., .apk or .so files)
- Static-Analysis on scale:
  - AndroGuard
  - Exodus-Core
  - Quark-Engine
  - APKiD
  - APKLeaks
  - MobSFScan
  - APKscan
  - FlowDroid
  - …

Image source: https://github.com/androguard/androguard

Image source: https://reports.exodus-privacy.eu.org/en/

Case study using FMD

- 5,728 firmware samples for experimenting with 75,141 unique apps
- We used firmware from seven vendors
- Static analysis
- We analysed the permissions and trackers used
- Dataset is free of charge available for researchers

ANDROID FIRMWARE DATASETS WITH NUMBER OF SAMPLES PER ANDROID VERSION.

ROM Vendor	# Firmware	# Apps	# Unique Packages	# Unique Apps (SHA-256)	v0	v4	v5	v6	v7	v8	v9	v10	v11
Google	1,023	169,392 (19.92%)	652 (0.38%)	37,255 (21.99%)	530	0	0	0	34	68	102	156	133
GrapheneOS	18	3,597 (0.42%)	243 (6.67%)	2,131 (59.24%)	0	0	0	0	0	0	0	0	18
Paranoid	99	19,332 (2.27%)	322 (1.67%)	1,616 (8.36%)	0	0	0	0	0	0	0	99	0
Carbon	1,300	68,901 (8.10%)	317 (0.46%)	2,725 (3.95%)	0	0	0	0	0	94	386	820	0
LineageOS	644	122,838 (14.44%)	457 (0.37%)	4,620 (3.76%)	0	0	0	0	0	0	0	200	444
OmniROM	2,107	292,879 (34.43%)	503 (0.17%)	15,815 (5.40%)	932	51	6	9	252	3	124	530	200
RROS	537	173,616 (20.41%)	688 (0.4%)	10,979 (6.32%)	0	0	0	0	0	0	0	537	0
Total	5,728	850,555	3,182 (0.37%)	75,141 (8.83%)	1,462	51	6	9	286	165	612	2,342	795
					25.52%	0.89%	0.10%	0.16%	4.99%	2.88%	10.68%	40.89%	13.88%

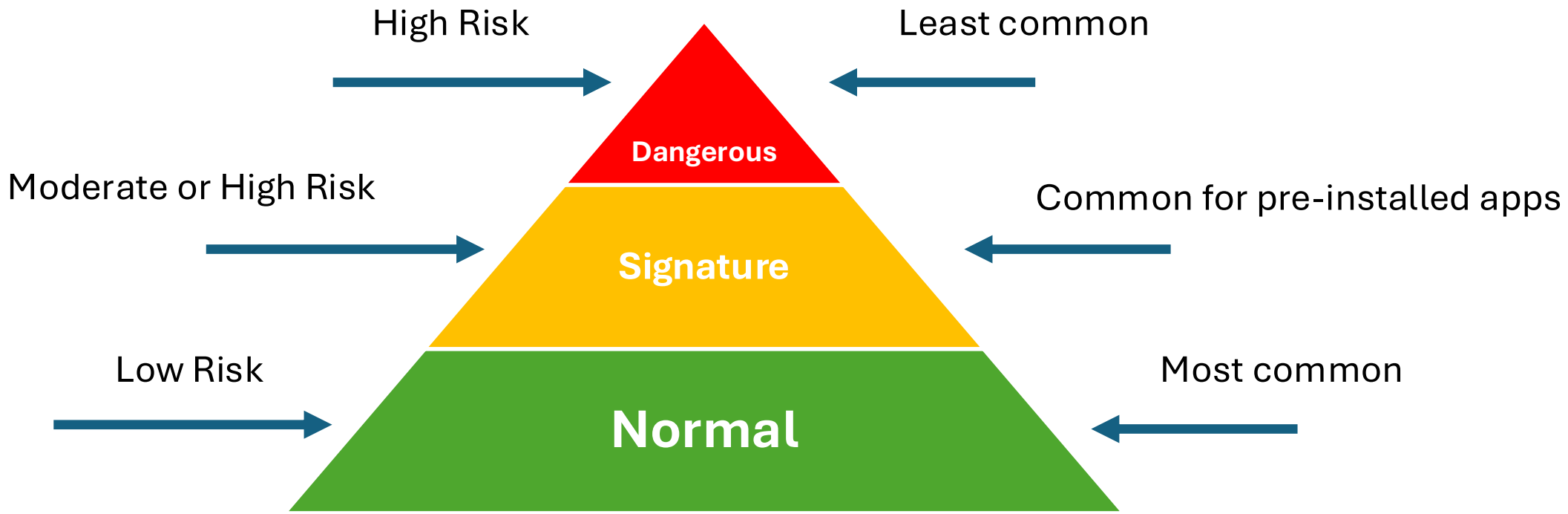
Table from our paper: https://ieeexplore.ieee.org/document/10172951

Permissions

Have you seen a consent or permission message when opening any of the pre-installed apps on your phone?

Usually, the user's consent is given by the first use of the device. Some apps ask for permissions, some don’t.

There are three main permission levels on Android



Permission analysis

Top 20 requested dangerous permissions for pre-installed apps on Android 10 and 11

# Rank	Permission	RROS V10	Paranoid v10	OmniROM v11	OmniROM v10	LineageOS v11	LineageOS v10	GrapheneOS v11	Google V11	Google v10	Carbon v10
1	android.permission.WRITE_EXTERNAL_STORAGE	16,512	2,895	4,755	12,676	11,652	5,251	378	5,305	6,218	6,142
2	android.permission.READ_EXTERNAL_STORAGE	12,682	2,680	4,486	11,871	8,837	3,701	378	4,258	4,948	5,746
3	android.permission.READ_PHONE_STATE	10,586	2,024	2,699	8,868	7,639	3,695	280	5,545	6,584	3,874
4	android.permission.READ_CONTACTS	9,224	2,270	2,771	8,005	6,664	3,112	302	4,284	5,332	4,008
5	android.permission.ACCESS_FINE_LOCATION	7,273	1,813	3,164	6,806	6,729	2,209	262	4,649	4,946	3,388
6	android.permission.GET_ACCOUNTS	7,890	1,877	2,171	6,592	4,960	2,666	248	4,234	5,328	3,218
7	android.permission.ACCESS_COARSE_LOCATION	6,139	1,835	2,042	5,823	4,490	1,800	234	4,556	5,088	2,929
8	android.permission.WRITE_CONTACTS	6,137	1,482	1,651	4,692	4,042	2,066	176	2,394	2,992	2,356
9	android.permission.CALL_PHONE	5,314	1,336	1,720	4,640	4,247	1,786	162	2,654	3,304	2,636
10	android.permission.CAMERA	5,055	1,112	1,498	4,203	3,043	1,152	180	2,588	3,072	1,890
11	android.permission.RECORD_AUDIO	4,721	1,129	1,182	3,026	3,760	1,531	126	2,698	3,234	1,215
12	android.permission.READ_CALL_LOG	3,615	940	1,399	3,248	3,169	1,209	126	1,463	1,782	1,767
13	android.permission.SEND_SMS	3,630	846	1,198	3,249	2,761	1,218	108	1,735	2,222	1,724
14	android.permission.READ_SMS	3,169	846	1,036	2,720	2,761	1,009	108	1,723	2,108	1,408
15	android.permission.WRITE_CALL_LOG	2,431	651	998	2,190	2,256	809	90	1,197	1,588	1,135
16	android.permission.READ_CALENDAR	2,925	652	600	1,989	1,332	1,000	54	931	1,248	862
17	android.permission.WRITE_CALENDAR	2,697	468	600	1,916	1,332	1,000	54	665	780	862
18	android.permission.ACCESS_BACKGROUND_LOCATION	1,519	273	1,073	1,404	1,437	200	54	2,017	1,502	273
19	android.permission.PROCESS_OUTGOING_CALLS	1,422	395	420	1,358	1,040	446	36	1,064	1,328	631
20	com.android.voicemail.permission.ADD_VOICEMAIL	1,200	281	400	1,058	884	400	36	532	690	589

Table from our paper: https://ieeexplore.ieee.org/document/10172951

88.14% of the permissions are signature-, 3.56% are dangerous-, and 8.21% are normal declared permissions.

28.57% of the permissions in our dataset are custom third-party permissions. We don't know what they do…

Tracking The Trackers

A tracker is a piece of software that logs the behaviour of a program or user.

Exodus: https://github.com/Exodus-Privacy/exodus-core

«εxodus analyses Android applications. It looks for embedded trackers and lists them. A tracker is a piece of software meant to collect data about you or what you do. In a way, εxodus reports are a way of knowing what really are the ingredients of the cake you are eating. εxodus does not decompile applications, its analysis technique is entirely legal.” – via https://exodus-privacy.eu.org/en/page/what/

Image source: https://reports.exodus-privacy.eu.org/en/

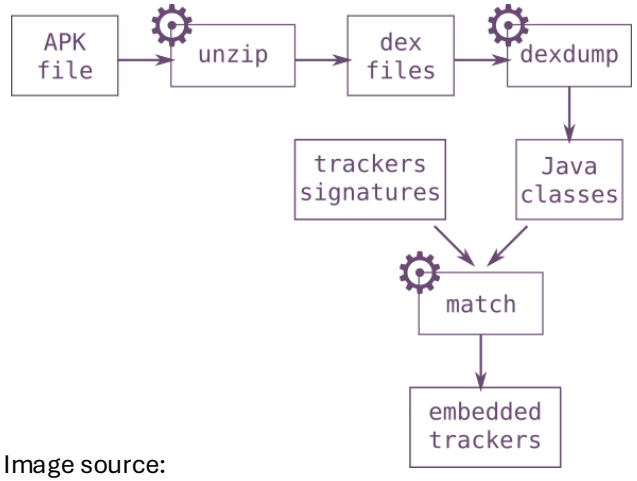


Image source: https://exodus-privacy.eu.org/en/post/exodus_static_analysis/

Java Class Matching:

- Benefits:
  - Small False Positive Rate
- Downsides:
  - If the tracker is unkown it will miss it

Tracker Results

- We found 41,175 known trackers with Exodus. Note: One app can have more than one tracker

- In total 24,597 (**20.53%**) of the pre-installed apps in our dataset use trackers.

A tracker is a piece of software that logs the behaviour of a program or user.

Are these trackers bad? Do they consent? This is still an open research question…

FirmwareDroid: Towards Automated Static Analysis of Pre-Installed Android Apps

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***Abstract*—Supply chain attacks are an evolving threat to the IoT and mobile landscape. Recent malware findings have shown that even sizeable mobile phone vendors cannot defend their operating systems fully against pre-installed malware. Detecting and mitigating malware and software vulnerabilities on Android firmware is a challenging task requiring expertise in Android internals, such as customised firmware formats. Moreover, as users cannot choose what software is pre-installed on their devices, there is a fundamental lack of transparency and control. To make Android firmware analysis more accessible and regain some transparency, we present FirmwareDroid, a novel open-source security framework for Android firmware analysis that automates the extraction and analysis of pre-installed software.**

FirmwareDroid streamlines the process of software extraction from Android firmware for static security and privacy assessments. With FirmwareDroid, we lay the groundwork for researchers to automate the security assessment of Android firmware at scale, and we demonstrated the capabilities of FirmwareDroid by analysing 5,728 Android firmware samples from various vendors. We analysed 75,141 unique pre-installed Android applications to study how common advertising tracker libraries (a piece of software that collects user usage data) are used and which permissions pre-installed Android apps inherit. We conclude that 20.53% of all apps in our dataset include advertising trackers and that 88.14% of all used permissions are signature-based.

***Index Terms*—Android Firmware, Pre-Installed Apps, Static Analysis, Security, Vulnerability**

I. Introduction

When we buy a smartphone or IoT device, we buy a piece of hardware and the pre-installed software on the hardware. If the device is part of the Android ecosystem, this software is called firmware, or Android firmware (ROM). If the device manufacturer provides it for reloading onto the device, it is usually in the form of a so-called firmware image. Android firmware consists of the Android operating system with its

just like regular apps downloaded and installed by users after the device is shipped, these pre-installed apps also potentially pose a security risk. On the one hand, they can contain vulnerabilities that attackers can exploit. On the other hand, they could have been intentionally equipped with functionality to spy on users or harm them in other ways. This is even more of a problem because pre-installed apps often have higher permissions than regular apps, cannot be easily removed by users, and, in contrast to apps from app stores, there are also no mechanisms like commenting, rating, and reporting potentially problematic apps.

Various well-documented incidents show that both security risks are not only theoretical. The risk of malware introduced by members of the supply chain is demonstrated, for example, by the malware families known as Chamois [1], [2], [3] and Triada [4], [5]. According to these sources, these have infected several million devices and remained undetected for months. In addition to these security risks, privacy risks from advertising trackers contained in the pre-installed apps are moving into the spotlight of the media [6]. Unfortunately, there are hardly any studies on how prevalent such advertising trackers are used in pre-installed apps nowadays.

Indications that this could be a problem are provided by the study of Gamba et al. [7]. It shows that a worrying number of companies known for their aggressive advertising strategies are associated with providers of Android firmware components. However, whether this complies with privacy regulations such as the GDPR or the CCPA remains unclear in the study. On the other hand, some providers of alternative Android firmware, such as LineageOS [8] and GrapheneOS [9], see in the commitment to a privacy-friendly Android firmware and the renunciation of so-called bloatware at least a market gap and opportunity. If we look at all of these risks and take the

Are we done yet?

Improve Tooling

- Most tools are far away from giving precise results:
  - Usually, high amounts of false positives
    - Vulnerabilities are often not exploitable
      - Lacking context awareness
      - Lack of configuration possibilities
- Re-Inventing the wheel
- Many tools do not scale well

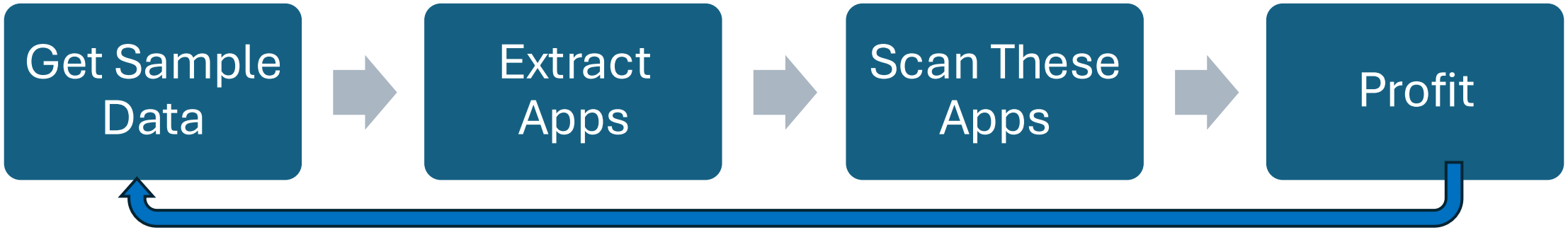
- **However, existing tools are used in research studies**
  - For instance, “AndroGuard” is use it many studies
  - Tools are used for comparison to develop new and better methods
  - Research tools are often unmaintained.
    - Thus, hard to setup after some time

State February 2025

- AndroGuard
- Quark-Engine
- APKiD
- Exodus-Core
- APKLeaks
- MobSFScan
- APKscan
- FlowDroid
- Trueseeing
- APKScan
- Deprecated:
  - Qark (deprecated, no updates by the author)
  - Androwarn (deprecated, no updates by the author)
  - SUPER Android Analyzer (deprecated, discontinued by the author)

- **YOUR TOOL HERE?**

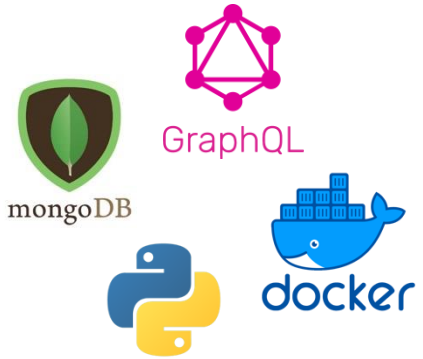
Moving Forward



Improve and repeat the experiments

Improvements from the original case study:

- Refactored the whole code base
  - Reimplemented the extraction process:
    - Better routines
    - More vendors supported
    - Added support for Android custom file formats
  - Added more tools
- Maybe program a GUI



Next Goal: Making the dynamic analysis of pre-installed apps possible

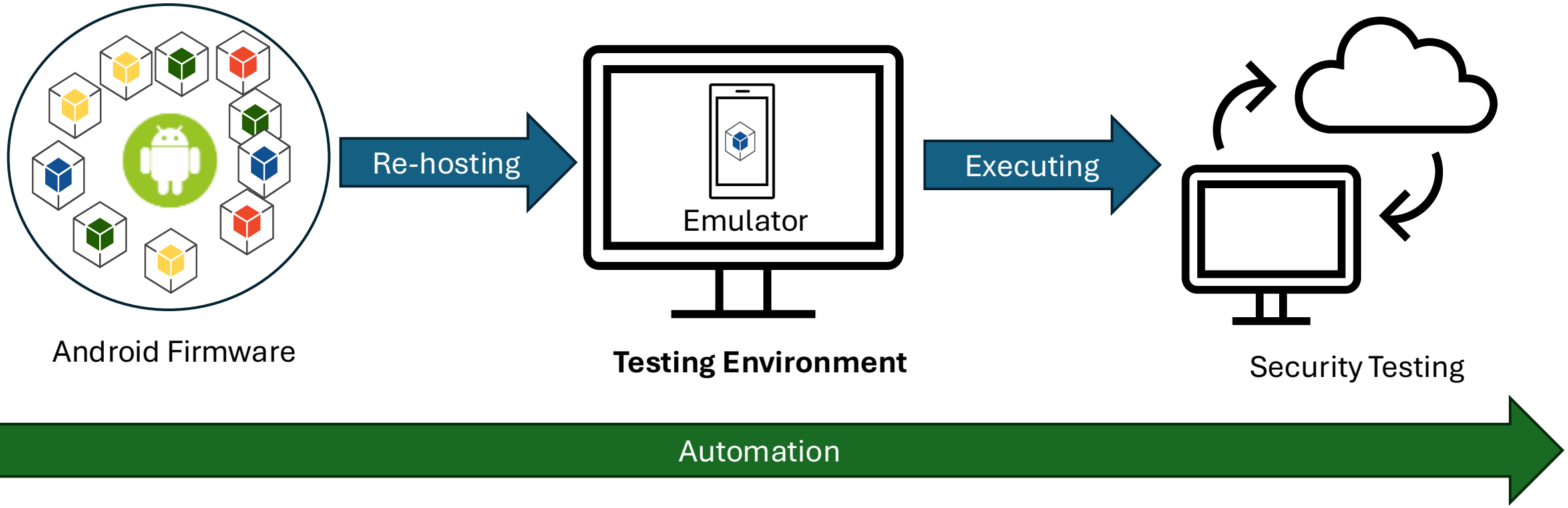
Challenges Testing Pre-Installed Apps

Static analysis is limited and can't give us full insights

1. Problematic Installation
   - App privileges
     - Signature: App must be signed with a platform key
     - Privilege: App must live in /system/priv-apps/ (System partition are read-only)
   - Singleton Apps: App can't be installed because it already exists in the testing environment
   - Resources must be loaded before the app is started
   - Apps must be installed during first boot process
2. App runs under a shared UID
3. App is headless
4. App is dependent on custom hardware
5. App has software dependencies to other components (Collusion)
   - Companion apps
   - Deamons
   - Binaries (ARM, x86)
   - Custom Android framework
   - Remote servers
6. App prevents execution
   - CPU Architecture (x86 / ARMv8a)
   - Environment checks (emulator detection)
   - Root detection

Re-Hosting Pre-Installed Apps

Objective: Re-hosting pre-installed apps from device firmware to the Android emulator




THANK
YOU

Q & A

Material:

- Paper:
  - *FirmwareDroid: Towards Automated Static Analysis of Pre-Installed Android Apps*
- Code:
  - https://github.com/FirmwareDroid/FirmwareDroid

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